

RESEARCH ARTICLE

WILEY

Developmental dyslexia and creativity: A meta-analysis

Nadyanna M. Majeed  | Andree Hartanto | Jacinth J. X. Tan 

School of Social Sciences, Singapore
Management University, Singapore

Correspondence

Nadyanna M. Majeed and Jacinth J. X. Tan,
School of Social Sciences, Singapore
Management University, Level 4, 90 Stamford
Road, Singapore 178903.
Email: nadyannam.2017@smu.edu.sg (N. M. M.)
and jacinthtan@smu.edu.sg (J. J. X. T.)

Despite difficulties in reading and writing, some research suggests that dyslexia may be related to higher levels of creativity. However, this pattern is not consistently observed. The current research sought to ascertain whether individuals with clinically diagnosed dyslexia exhibit higher creativity than controls through a meta-analysis. Fourteen studies that assessed the creativity of 397 individuals with clinically diagnosed dyslexia and 453 controls were reviewed. Random-effects meta-analysis revealed an overall non-significant difference in creativity scores between those with dyslexia and controls. Additionally, method factors such as the type of creativity task and whether intelligence was controlled for, as well as sample-related factors such as gender, did not explain differences in the dyslexia-creativity relationship. Nonetheless, individuals with dyslexia significantly outperformed controls in creativity scores in adult samples, but not in younger child/adolescent samples. Overall, the current findings provide limited support for the idea that individuals with dyslexia are more creative, and that past evidence of this relationship may be limited to adult samples.

KEYWORDS

creativity, dyslexia, meta-analysis

1 | INTRODUCTION

Developmental dyslexia is a major learning disability in children and adults. It is neurobiological in nature (Duffy, Denckla, Bartels, & Sandini, 1980; Ramus, 2004; B. A. Shaywitz et al., 2002) and its symptoms include persistent and substantial difficulties in reading and spelling (Lyon, Shaywitz, & Shaywitz, 2003). Globally, it is the most common

learning disability, affecting more than 80% of those identified as having a learning disability (Lerner, 1989). Developmental dyslexia exists in every language studied, although the severity and type of difficulties experienced by individuals with dyslexia may vary with the characteristics of the language (Büttner & Hasselhorn, 2011; Paulesu, 2001; Peterson & Pennington, 2012; Ziegler, Perry, Ma-Wyatt, Ladner, & Schulte-Körne, 2003).

Developmental dyslexia is typically associated with speed deficits in reading-related tasks, poor phonological and/or phonemic awareness and poor performance in rapid-naming tasks (Lum, Ullman, & Conti-Ramsden, 2013; O'Brien, Wolf, & Lovett, 2012; Wimmer, 1993). This includes difficulties in tasks such as identifying or generating rhyming words, or counting the number of syllables in words. As such, children with dyslexia often require targeted remedial instruction for reading and spelling from a young age (O'Hare, 2010). Even with targeted remedial instruction, these challenges persist into adolescence and adulthood, with most adults with dyslexia never achieving automatic reading (S. E. Shaywitz & Shaywitz, 2005; Yap & van der Leij, 1994). These difficulties translate to poor scores on tests of working memory and phonological coding (Jeffries & Everatt, 2004) and tests of phonological processing (Thomson, 2003; Whitehouse, 1983). Nevertheless, while individuals with dyslexia experience impairments in reading and related tasks, this does not necessarily translate to impairments in other domains. For instance, many individuals with dyslexia display high levels of non-verbal intelligence (Craggs, Sanchez, Kibby, Gilger, & Hynd, 2006; Das, Mishra, & Kirby, 1994) and faster visuo-spatial processing compared to individuals without dyslexia (von Károlyi, Winner, Gray, & Sherman, 2003).

Despite the typical verbal impairments, some research has suggested and found that individuals with dyslexia significantly outperform their peers without dyslexia on tests of creativity (e.g., Bigozzi, Tarchi, Pinto, & Donfrancesco, 2016; Chakravarty, 2009; Everatt, 1997; Everatt, Steffert, & Smythe, 1999; Tarver, Ellsworth, & Rounds, 1980). This finding highlighted a potential strength among those with dyslexia, beyond the more commonly known impairments that have contributed to considerable stigmatisation of these individuals (Alexander-Passe, 2015; Morris & Turnbull, 2007). If creativity is indeed enhanced in individuals with dyslexia, one important implication is that encouraging these individuals to build on this strength can potentially improve their self-esteem and overall emotional adjustment (Gunnel Ingesson, 2007). Paying greater attention to this unique strength may also help to change public perceptions of dyslexia as largely a disability, thereby reducing the stigma against individuals with dyslexia (Cockcroft & Hartgill, 2004; Tafti, Hameedy, & Baghal, 2009).

Two key explanations have been proposed for why individuals with dyslexia may be more creative. First, the *compensation hypothesis* posits that individuals with dyslexia undergo compensatory neurological development in other brain structures to counteract deficits caused by dyslexia, such as the left anterior inferior frontal gyrus and the lateral frontopolar cortex, both of which have been associated with creativity (Abraham et al., 2012; Galaburda, Sherman, Rosen, Aboitiz, & Geschwind, 1985; Geschwind & Galaburda, 1987; Winner et al., 2001). The *early choice hypothesis*, on the other hand, suggests that higher levels of creativity among individuals with dyslexia arise due to practice effects that stem from adopting non-typical methods early on, such as using pictures rather than words, to process information (Bigozzi et al., 2016; Tafti et al., 2009).

However, a number of more recent studies have revealed conflicting findings. In one set of studies, an opposite effect was found such that individuals with dyslexia were less creative than their peers without dyslexia (e.g., Kasirer & Mashal, 2017). This was explained by the *facilitation hypothesis*, which suggested that fluency in written language facilitates creativity through broadening the range of experiences that one has (Lee & Kim, 2011; Ritchie, Luciano, Hansell, Wright, & Bates, 2013; Wang, 2012). As such, individuals with dyslexia, due to their language impairments, lack such experiences that provide foundational resources for creativity (Sturgell, 2008). In another set of studies, no significant differences were found between creativity scores of groups with and without dyslexia (e.g., Cancer, Manzoli, & Antonietti, 2016; Łockiewicz, Bogdanowicz, & Bogdanowicz, 2014). These mixed findings raised questions about whether the effect of dyslexia on creativity truly exists.

A major challenge in this area of work is that the available studies in the literature that differ in their conclusions also differ vastly in their methodologies, so the results across these studies are not necessarily comparable. One key methodological difference among previous studies is the use of verbal or non-verbal creativity tests. Because

individuals with dyslexia have impaired phonological and reading skills, they are inherently disadvantaged by creativity tests that are verbal in nature, so their creativity scores may not reflect their actual creativity (e.g., Kasirer & Mashal, 2017). Research has shown that verbal tasks tend to disadvantage children with learning disabilities in general, including those with developmental dyslexia, resulting in their underperformance in these verbal tasks compared to non-disabled peers, even if both groups produced comparable scores on non-verbal tasks (Eisen, 1989; Tarver et al., 1980; Wallach, 1970). In fact, some studies that used only non-verbal tasks revealed an advantage for the group with dyslexia (e.g., Everatt et al., 1999; Pačalska, Bogdanowicz, Tomaszewska, Łockiewicz, & Bogdanowicz, 2009), although this was, again, not consistently found (e.g., Alves & de Cássia Nakano, 2014; Everatt et al., 1999).

A second methodological difference among previous studies is that the participants with dyslexia and controls were not always matched on intelligence. The use of intelligence-matched controls is important. On average, individuals with dyslexia tend to have lower intelligence scores than their peers without dyslexia (Gunnel Ingesson, 2006; van Bergen et al., 2014). Additionally, some evidence suggests that intelligence is positively correlated with creativity (Furnham, Batey, Anand, & Manfield, 2008; Preckel, Holling, & Wiese, 2006; Schubert, 1973). More recent work has posited that intelligence and creativity are associated due to the similar executive processes involved (Khalil, Godde, & Karim, 2019; Silvia, 2015). For example, *updating* ability, which refers to the monitoring of incoming information and the revision of obsolete information in the working memory, predicts both intelligence and creativity (Benedek, Jauk, Sommer, Arendasy, & Neubauer, 2014). Despite these known associations between intelligence and creativity, many studies that examined the dyslexia-creativity relationship did not control for or match participants on intelligence (e.g., Kasirer & Mashal, 2017; Pačalska et al., 2009). Therefore, the observed differences in creativity between those with dyslexia and those without dyslexia may actually be due to differences in intelligence.

1.1 | Summary of goals of the meta-analysis

Given that identifying enhanced creativity as a strength among those with dyslexia has important implications for improving their psychological well-being, as well as for addressing public perceptions related to stigma, a critical evaluation of the mixed evidence in the current literature is warranted. The primary goal of this research was to conduct a systematic integration and review of the current empirical findings of the relationship between dyslexia and creativity and estimate an overall effect size of the relationship through a meta-analysis. A positive and significant meta-analytic effect size would support the notion that dyslexia is linked to higher creativity. A secondary goal was to examine potential methodological factors as moderators of the relationship, which may account for the mixed findings in the literature. Based on the methodological issues outlined earlier, three specific moderators were examined: (a) the type of creativity task used (verbal or non-verbal), (b) the type of creativity outcome assessed (e.g., fluency, originality), and (c) whether the dyslexia and control groups were matched on intelligence (matched vs. non-matched). These moderator tests would help identify the specific contexts in which individuals with dyslexia may perform better, or worse, than their peers without dyslexia.

The current meta-analytic investigation of the existing dyslexia and creativity research is important in three ways. First, a meta-analysis aggregates effect sizes across different studies by taking into account differences in study sample sizes, providing a more powerful test that yields a more precise effect size estimate of the overall dyslexia-creativity relationship. Second, by coding for methodological and sample characteristics of the studies, it allows for a systematic investigation of whether method or sample factors may explain differences in the effect sizes and potentially account for some of the mixed patterns found in the literature. Finally, a comprehensive synthesis of the current studies that highlights method and sample characteristics provides a useful guide for future empirical work in the area by highlighting critical gaps in the literature and the factors that are important to consider in future study designs. More broadly, the current meta-analysis aims to motivate better empirical work in the area, which is necessary for informing and refining existing theories of the dyslexia and creativity relationship.

It is important to note that many studies in the existing literature that purport to study the effects of dyslexia have used assessments of reading ability, such as reading comprehension or reading speed, as indicators of dyslexia, rather than using formal clinical diagnoses. However, compared to assessments of reading ability, the clinical diagnosis of dyslexia involves a more extensive clinical review of the individual's developmental, medical, educational, and family history, reports of test scores and teacher observations, and response to academic interventions. Importantly, the diagnosis requires that the individual's reading difficulties must not be better explained by developmental, neurological, sensory, or motor disorders and must significantly interfere with academic achievement, occupational performance, or activities of daily living (American Psychiatric Association, 2013), which are not distinguished in a crude assessment of reading ability. Additionally, recent research has found that individuals with dyslexia can also be found at the higher end of the reading spectrum, and that samples of poor readers are often mainly composed of individuals without dyslexia (Wagner et al., 2020). Therefore, to better isolate the effects of dyslexia on creativity, we limited our review to studies that recruited samples with a formal clinical diagnosis of dyslexia, such as by a clinical psychologist or an educational psychologist.

2 | METHOD

2.1 | Inclusion criteria

A literature search was conducted in *ERIC*, *PsycINFO*, *PubMed*, *Scopus* and *Web of Science* using the following keywords: (dyslexi* OR "reading disab*" OR "reading impair*" OR "reading disorder") AND (creativ* OR "divergent thinking") for all reports available by June 2020. Manual searches were conducted in *Annals of Dyslexia*, *Dyslexia*, *Google Scholar*, and *Reading and Writing*. As no previous meta-analyses related to dyslexia and creativity were found, these searches were also supplemented by examining the reference sections of past journal articles on the topic of dyslexia and creativity. To capture unpublished literature, a manual search was conducted in *OpenGrey*, a European database including doctoral dissertations and conference papers, and *ProQuest Dissertations & Theses*. Additionally, requests for unpublished, dissertation, under-review and in press data were sent to individual researchers in the field of dyslexia via email.

Figure 1 shows the PRISMA flowchart that outlines the overall process for selecting studies for inclusion in this meta-analysis (Moher, Liberati, Tetzlaff, Altman, & the PRISMA Group, 2009). In total, the search resulted in 473 potentially eligible records. After removing duplicates, 326 potentially eligible records were screened for inclusion based on abstracts, following which 50 full-text records were screened for inclusion in the current meta-analysis based on the following inclusion criteria:

1. Studies involving developmental dyslexia were included as long as they reported using a clinical diagnosis, for example, diagnosis by educational or clinical psychologists, or by medical centres or specialists.
2. Studies were included if creativity was explicitly studied through a creativity task, such as a divergent thinking task.
3. Studies were included if information necessary to compute effect sizes were reported.
4. If a study was eligible but did not report the appropriate statistics, original authors of the study were contacted directly to obtain usable data. Out of the seven authors contacted, six of them provided the requested data. The remaining author no longer had the original data and hence was unable to provide the appropriate statistics.

Studies that defined dyslexia based on cut-off points, such as reading comprehension at one *SD* below the mean, was excluded. Based on a further examination of the potentially eligible records, 10 records met inclusion criteria 1 to 3. Of these records, 9 (0% unpublished) met all inclusion criteria and were used in the meta-analysis, providing a total of 14 independent samples. The 14 samples included a total *N* of 850 (*Mdn* = 39.5, *M* = 60.71, *SD* = 53.41). Based on available reports, the age range of the samples was 7–55 years (*M* = 16.31, *SD* = 6.40) and there was an average gender proportion of 41.14% females (*SD* = 10.72%).

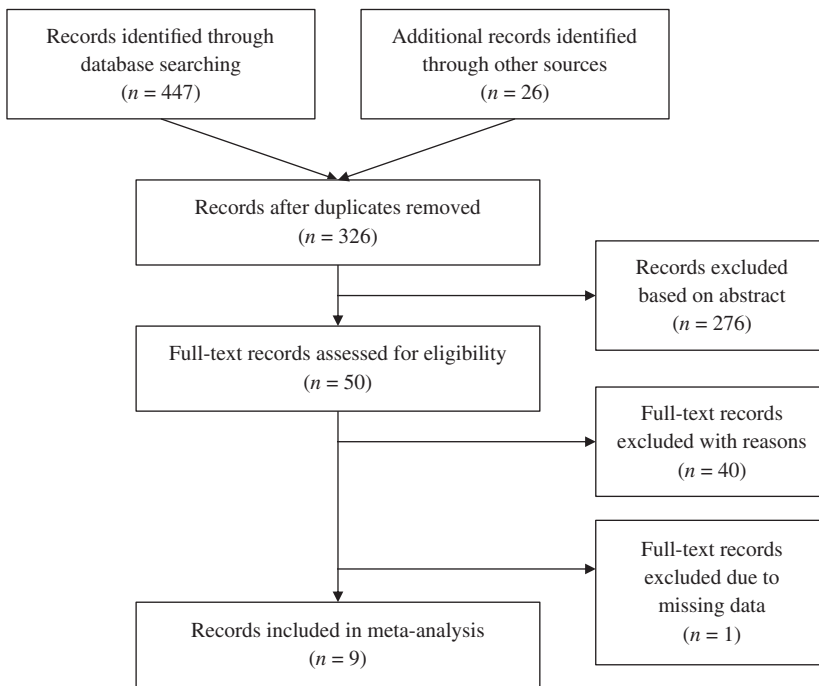


FIGURE 1 PRISMA flowchart

2.2 | Coding of study characteristics

The following study characteristics were coded: (a) publication source (journal article, unpublished data, dissertation, conference paper), (b) country where the study was conducted, (c) whether intelligence was controlled for or measured, and if yes, the intelligence measure used and the respective intelligence scores of each group, (d) type of creativity task used (verbal or non-verbal), and (e) type of creativity outcome assessed (e.g., fluency, flexibility, originality, elaboration).

For the creativity assessment, the mean and SD of scores achieved by both groups (with dyslexia and without dyslexia) were recorded whenever available. Available demographic information such as mean age and gender composition were also coded. All information was obtained directly from the Method sections of the respective studies, tables of descriptive statistics provided in the articles, or authors who responded to e-mail requests.

Study characteristics were coded independently by two coders (the first author and a trained research assistant) who then discussed and resolved discrepancies after the initial coding process. The agreement for all variables was good. For categorical variables, an average of $\kappa = 0.93$ ($\kappa_{\text{range}} = 0.79$ to 1.00) was obtained. For non-categorical variables, an average of $\kappa = 0.96$ ($\kappa_{\text{range}} = 0.71$ to 1.00) was obtained.

2.3 | Meta-analytic approach

2.3.1 | Effect size index

The effect size index used was Hedges' standardised g , a bias-corrected estimate of Cohen's standardised d which represents the difference between two group means divided by their pooled SD (Cohen, 1988; Hedges, 1981). Effect sizes were calculated such that positive values indicated an effect consistent with the hypothesis that individuals with dyslexia produce higher creativity scores.

This calculation reflects the proposed explanations in the literature previously mentioned (the *compensation hypothesis* and the *early choice hypothesis*) that individuals with dyslexia would outperform individuals without dyslexia on tests of creativity. Negative effect sizes would indicate that individuals with dyslexia would produce lower creativity scores than controls, consistent with the *facilitation hypothesis*.

Six samples (43%) provided multiple effect sizes of interest. This violates the statistical assumption that effect sizes are independent. For samples that reported multiple scores for various creativity constructs (e.g., *Widening score*, *Connecting score*, and *Reorganizing score* from the *Widening, Connecting, and Reorganizing test* in Cancer et al., 2016), an aggregated effect size within that sample was manually calculated. Hence, for the overall effect size estimates and tests of heterogeneity, each independent sample had only one effect size.

However, in the moderator analysis of the type of creativity task (i.e., verbal task vs. non-verbal task), scores on verbal and non-verbal tasks were not aggregated into a combined “mixed task” score. Instead, scores were aggregated within each type of task, such that if a sample used both verbal and non-verbal tasks, it would contribute two effect sizes instead of one. There were three samples (21%) in which this occurred. Additionally, there was one sample (7%) in which a verbal score could be isolated, but a non-verbal score could not as no purely non-verbal subtests were used.

2.3.2 | Data analysis

To investigate the primary goal of whether overall significant differences in creativity scores exist between the groups with and without dyslexia, an overall meta-analytic effect size was computed. Moderator analyses were conducted by examining whether the magnitude of differences between the groups with and without dyslexia differed significantly as a function of a given moderator.

The overall effect size estimates and heterogeneity tests of effect sizes for all samples, as well as a test of publication bias, were conducted in R version 3.6.3 (R Core Team, 2020) using the meta-analytic software metafor version 2.4-0 (Viechtbauer, 2010). As the test for heterogeneity was significant, $Q = 43.54$, $p < .001$ (for $k = 14$, $df = 13$), the overall meta-analytic effect size was estimated using random-effects, while the moderator tests were conducted using mixed-effects analyses. In both types of analysis, a restricted maximum-likelihood estimation was used.

3 | RESULTS

3.1 | Overall effect sizes

The effect sizes of the associations between developmental dyslexia and creativity are listed in Table 1 and Figure 2. The overall standardised mean difference effect size ($k = 14$) was 0.13 with 95% CI $[-0.17, 0.43]$, $z = 0.85$, $p = .40$. This suggests that groups with dyslexia did not outperform groups without dyslexia on tests of creativity.

3.2 | Test of publication bias

A test of publication bias was conducted to investigate potential threats to the validity of the overall meta-analytic effect sizes. A funnel plot was created, using metafor, to analyse the distribution of effect sizes in the samples to determine potential biases in study inclusion. The goal of the funnel plot is to assess whether the overall effect size estimate in the meta-analysis is inflated due to the exclusion of studies where the null hypothesis was not rejected (Sterne & Egger, 2001). A symmetrical distribution of effect sizes in the funnel plot suggests that the effect size estimate is likely to be less biased. Figure 3 represents the funnel plot for all samples.

TABLE 1 Effect sizes, sample sizes and moderator codes for each sample in the meta-analysis

Paper	Published	N	Age range	Mean age	Country	Proportion of female	Intelligence matched	Creativity measure	Creativity measure (type)	Overall Hedge's g	Non-verbal Hedges' g	Verbal Hedges' g
Alves and de Cássia Nakano (2014)	Yes	26	9–11 (majority)	10.77	Brazil	0.54	Yes	CFCT	Non-verbal	–0.15	–0.15	
Bigozzi et al. (2016)	Yes	190	8.1–11.6	9.42	Italy	0.29	Not reported	CAP	Mixed	0.41		
Cancer et al. (2016)	Yes	52	12–15	Not reported	Italy	Not reported	Not reported	WCR	Mixed	0.08		–0.33
Everatt (1997)	Yes	62	18–55	Not reported	England	0.33	Yes	AUT & DT	Mixed	0.62	0.68	0.56
Everatt et al. (1999)	Yes	36	18–44	24.03	England	0.44	Yes	AUT & PPT	Mixed	0.71	0.76	0.66
Everatt et al. (1999)	Yes	39	18–53	24.77	England	0.41	Yes	FP	Non-verbal	0.69	0.69	
Everatt et al. (1999)	Yes	33	7–9	7.90	England	0.36	Yes	PPT	Non-verbal	–0.23	–0.23	
Everatt et al. (1999)	Yes	45	11–13	12.47	England	0.31	Yes	PPT	Non-verbal	–0.21	–0.21	
Kapoula et al. (2016)	Yes	41	12–15	13.30	Belgium	0.51	Yes	TTCT-F	Non-verbal	0.58	0.58	
Kasirer and Mashal (2017)	Yes	37	9–11	11.20	Israel	0.32	Yes	NMGT	Verbal	–0.52		–0.52
Kasirer and Mashal (2017)	Yes	35	13–16	14.29	Israel	0.29	No	NMGT	Verbal	–1.41		–1.41
Kasirer and Mashal (2017)	Yes	34	18–25	22.82	Israel	0.56	Yes	NMGT	Verbal	0.68		0.68
Łockiewicz et al. (2014)	Yes	180	18–30	22.00	Poland	0.56	Yes	VFT & TCT	Mixed	–0.10	–0.02	–0.26
Pachalska et al. (2009)	Yes	40	Not reported	22.75	Poland	Not reported	No	FEATS	Non-verbal	0.66	0.66	

Abbreviations: AUT, Alternative Uses Task (Guilford & Hoepfner, 1971); CAP, Creativity Assessment Packet – Divergent Thinking Subtest (Williams, 1993); CFCT, Child Figural Creativity Test (Nakano, Wechsler, & Primi, 2011); DT, Drawing Task (Everatt, 1997); FEATS, The Formal Elements of Art Therapy Scale (Gantt & Anderson, 2009); FP, Five Problems (Finke, Ward, & Smith, 1996; Metcalfe, 1986; Weisberg & Alba, 1981); NMGT, Novel Metaphor Generation Task (Kasirer & Mashal, 2017); PPT, Picture Production Task (Everatt et al., 1999); TCT, Urban–Jellen Test for Creative Thinking–Drawing Production–Polish adaptation (Malczak, Jaworowska, & Starczak, 2000); TTCT-F, Torrance Tests of Creative Thinking – Figural Form (Torrance, 1966); VFT, Verbal Fluency Tasks (Łockiewicz et al., 2014); WCR, Widening, Connecting and Reorganizing Test (Antonietti, Giorgetti, & Pizzigalli, 2011).

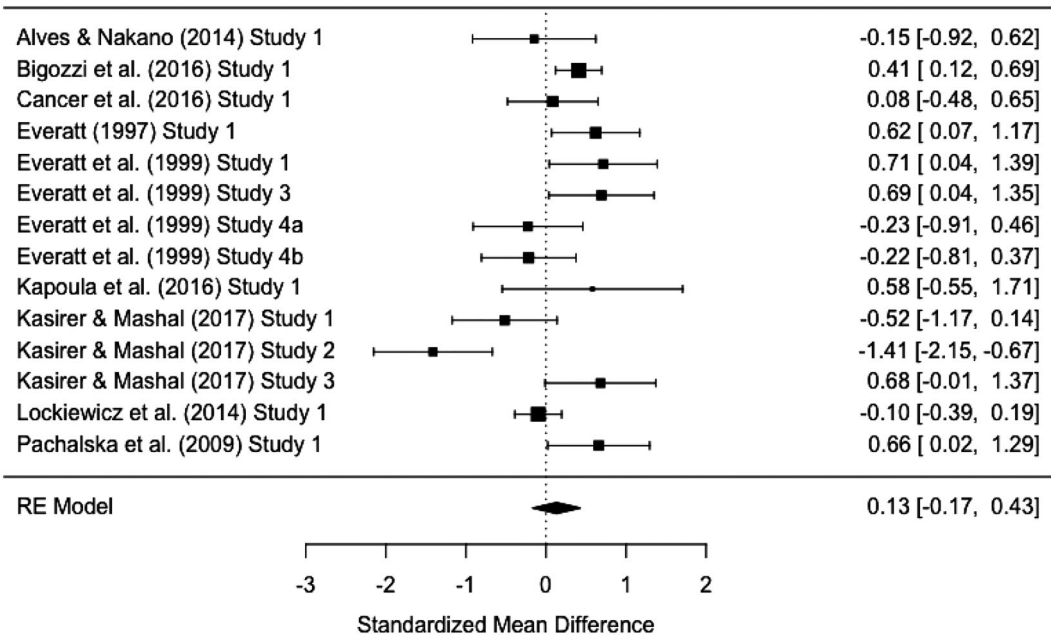


FIGURE 2 Effect sizes and confidence intervals for all samples

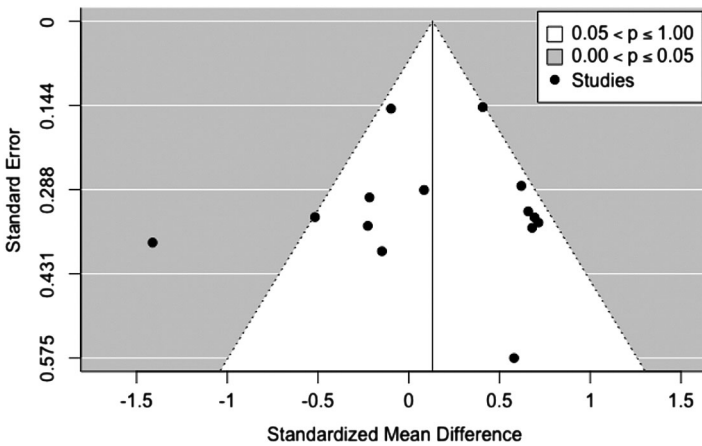


FIGURE 3 Funnel plot for distribution of effect sizes

To test for funnel plot asymmetry, a rank correlation test was conducted. The rank correlation test involves calculating Kendall's τ , a non-parametric correlation of effect sizes and their SEs (Kendall, 1938). A significant correlation produced from this test would suggest the likelihood of inclusion bias in the samples. For all samples, $\tau = -0.19$, $p = .39$. As the rank correlation test was non-significant, inclusion bias was not likely in the samples.

3.3 | Moderators of the effect sizes

3.3.1 | Verbal versus non-verbal creativity task

We tested if the effect size was significantly different between non-verbal tasks and verbal tasks using the mixed-effects analysis. Where possible, samples that contributed scores for both verbal and non-verbal tasks had their

TABLE 2 Analysis of categorical moderators of the dyslexia–creativity relation

Moderator	Level	N	k	g	SE _g	95% CI _{LB}	95% CI _{UB}
Creativity measure (type)	Non-verbal	502	9	0.28	0.15	−0.02	0.57
	Verbal	436	7	−0.08	0.28	−0.63	0.46
Creativity measure (mixture)	Mixed	468	4	0.35	0.19	−0.02	0.72
	Non-mixed	382	10	0.008	0.21	−0.40	0.42
Creativity outcome	Fluency	406	7	0.11	0.18	−0.24	0.47
	Originality	398	7	0.03	0.27	−0.51	0.57
Intelligence matched	Yes	533	10	0.18	0.15	−0.12	0.48
	No	75	2	−0.37	1.03	−2.40	1.66
Mean age	Children and adolescents	459	8	−0.18	0.21	−0.58	0.23
	Adults	391	6	0.48	0.18	0.13	0.82

Abbreviations: g, Meta-analytic effect size (Hedges' g); k, Number of samples; N, Sample size; SE_g, Standard error of effect size; 95% CI_{LB} and CI_{UB}, 95% Confidence interval.

effect sizes split into a verbal effect size and a non-verbal effect size ($k = 4$). Studies that did not report separate verbal and non-verbal scores or effect sizes were removed from the analysis ($k = 1$). The overall standardised mean difference non-verbal effect size ($k = 9$) was 0.28, with 95% CI [−0.02, 0.57], whereas the overall standardised mean difference verbal effect size ($k = 7$) was −0.08, with 95% CI [−0.63, 0.46]. As the two 95% CIs overlapped, and the test for the moderator was non-significant, $Q = 1.59$, $p = .21$ (for $k = 16$, $df = 1$), this suggests that the type of creativity task did not moderate the effect of dyslexia on creativity. These results are presented in Table 2 and Figure 4.

To compare the effects of being assessed via mixed creativity tasks (i.e., both verbal and non-verbal) versus non-mixed creativity tasks (i.e., purely verbal or non-verbal), we also tested if the effect sizes differed significantly between mixed tasks and non-mixed tasks. The overall standardised mean difference effect size for mixed tasks ($k = 4$) was 0.35, with 95% CI [−0.02, 0.72], whereas the overall standardised mean difference effect size for non-mixed tasks ($k = 10$) was 0.008, with 95% CI [−0.40, 0.42]. As the two 95% CIs overlapped, and the test for the moderator was non-significant, $Q = 1.27$, $p = .26$ (for $k = 14$, $df = 1$), this further suggests that the nature of the creativity tasks did not moderate the effect of dyslexia on creativity.

3.3.2 | Creativity assessed as fluency versus originality

Additionally, we conducted subgroup analyses based on the type of creativity outcome assessed. Creativity is frequently measured in terms of four common outcomes (Plucker & Makel, 2010): fluency (number of ideas generated), flexibility (number of categories of ideas generated), originality (number of unique ideas generated), and elaboration (level of detail of ideas generated). However, we were unable to conduct subgroup analyses for flexibility ($k = 2$) and elaboration ($k = 2$) due to the large heterogeneity and small number of samples available. The overall standardised mean difference fluency effect size ($k = 7$) was 0.11, with 95% CI [−0.24, 0.47], $z = 0.62$, $p = .54$, while the overall standardised mean difference originality effect size ($k = 7$) was 0.03, with 95% CI [−0.51, 0.57], $z = 0.11$, $p = .91$. These results suggest that groups with dyslexia did not significantly outperform groups without dyslexia in terms of fluency nor originality. These results are presented in Table 2.

3.3.3 | Intelligence matched versus not matched

We also examined whether the effect size of dyslexia on creativity would differ between samples that used intelligence-matched controls versus not. Studies were included in the analysis if they explicitly stated that

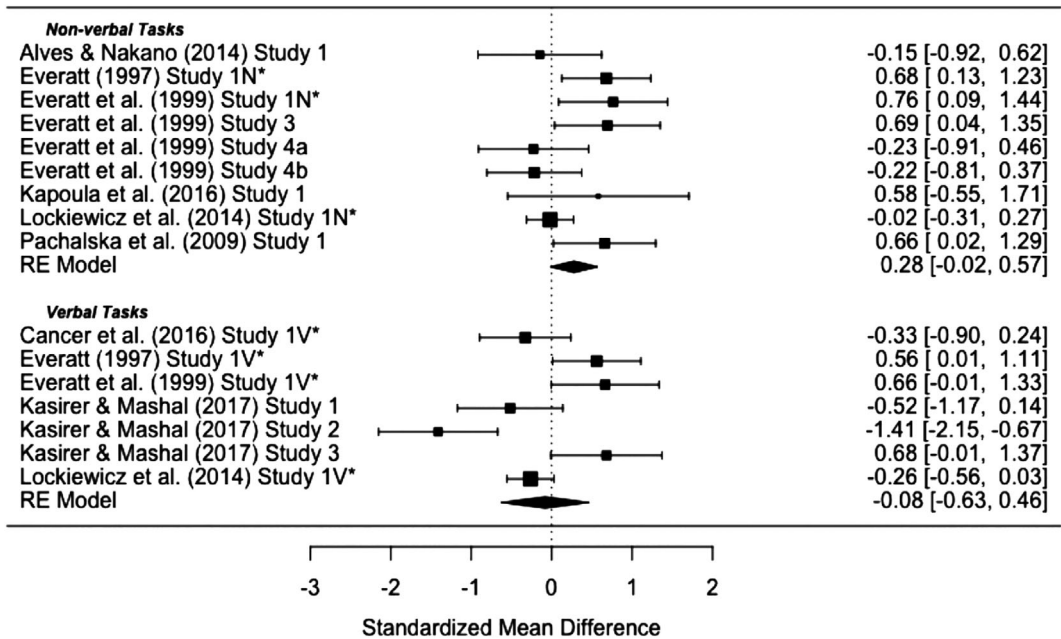


FIGURE 4 Effect sizes and confidence intervals for non-verbal and verbal tasks. Asterisks refer to samples with both verbal and non-verbal effect sizes

intelligence was matched or controlled for, or if they reported the appropriate statistics which allowed for comparison of intelligence scores between the two groups. The overall standardised mean difference effect size for matched samples ($k = 10$) was 0.18, with 95% CI $[-0.12, 0.48]$, whereas the overall standardised mean difference effect size for non-matched samples ($k = 2$) was -0.37 , with 95% CI $[-2.40, 1.66]$. As the two 95% CIs overlapped, and the test for the moderator was non-significant, $Q = 1.06$, $p = .30$ (for $k = 12$, $df = 1$), this suggests that the effect size of dyslexia on creativity did not differ significantly different between samples where intelligence was matched or controlled for versus not. These results are presented in Table 2.

3.3.4 | Differences in intelligence

To further examine the role of intelligence for dyslexia and creativity, we also tested if the dyslexia-creativity effect size was moderated by the strength of differences in intelligence between groups with and without dyslexia. For a subset ($k = 10$) of samples, where this information was available, we computed the standardised mean difference in intelligence between groups with and without dyslexia, which was entered as the moderator. The moderator test was non-significant, $b = 0.79$, 95% CI $[-0.28, 1.87]$, $z = 1.43$, $Q = 2.06$ (for $k = 10$, $df = 1$), $p = .15$, suggesting that the dyslexia-creativity effect size did not differ by differences in intelligence.

3.3.5 | Exploratory analyses

Two additional variables, mean age and proportion of females in the study, were explored as potential moderators of the dyslexia-creativity effect size. Age as a moderator was examined as a continuous variable, as well as a categorical variable with two levels. There was a non-significant trend for mean age as a continuous moderator variable,

$b = 0.05$, 95% CI $[-0.0039, 0.10]$, $z = 1.81$, $Q = 3.29$ (for $k = 12$, $df = 1$), $p = .07$, such that the dyslexia-creativity effect size increased with age. For mean age examined as a categorical moderator variable, the samples were split into two groups—children and adolescents (7–16) versus adults (18–55). Subgroup analyses revealed that for the group with children and adolescents ($k = 8$), the dyslexia-creativity effect size was non-significant, $g = -0.18$, 95% CI $[-0.58, 0.23]$, $z = -0.85$, $p = .39$. Interestingly, for the group with adults ($k = 6$), the dyslexia-creativity effect size was significant, $g = 0.48$, 95% CI $[0.13, 0.82]$, $z = 2.72$, $p = .007$, such that adults with dyslexia outperformed adults without dyslexia on creativity tasks. The patterns are shown in Figure 5.

The test for proportion of females as a moderator was non-significant, $b = 1.85$, 95% CI $[-1.56, 5.26]$, $z = 1.06$, $Q = 1.13$ (for $k = 12$, $df = 1$), $p = .29$, suggesting that the proportion of females in the study samples did not moderate the effect size of dyslexia on creativity. The results of all of the exploratory analyses are presented in Tables 2 and 3.

4 | DISCUSSION

The current investigation presents the first comprehensive analysis of data from studies of creativity between groups with and without developmental dyslexia in several different alphabetic languages. Potential moderators were also identified and explored in order to account for the conflicting results produced by previous studies of dyslexia and creativity. Overall, the meta-analytic results showed that there was no significant difference between groups with and without developmental dyslexia in their creativity scores.

The current findings suggest that there are no strong evidence supporting group differences in creativity as a function of dyslexia. This was the case even after taking into account several potential moderators, namely the type of creativity task (verbal vs. non-verbal), type of creativity outcome (fluency and originality), whether intelligence was matched, mean age and the proportion of females in the sample. As such, the mixed findings in the available literature are unlikely to be due to methodological differences or sample characteristics that were captured in the available studies.

The findings of the current study did not provide strong support for the *compensation hypothesis*, which predicted a positive dyslexia-creativity relationship, nor the *facilitation hypothesis*, which predicted a negative dyslexia-creativity relationship. Nonetheless, the exploratory analysis with age is, to some degree, suggestive of the *early choice hypothesis*—that higher levels of creativity among those with dyslexia arise due to practice effects from the use of non-typical methods to process information early in childhood (Bigozzi et al., 2016; Tafti et al., 2009). Indeed, some research in the general population has found that creativity can be trained (Ridley & Birney, 1967) and that practice effects can improve performance on different creativity tasks that tap into similar cognitive processes (Tan & Qu, 2015). Therefore, it is possible that the effect of adopting non-typical methods on creativity takes time to accumulate over the lifespan and manifests only later in adult groups when sufficient practice has occurred, but not in younger children, as observed in our age analysis. However, we note that the studies in this review are cross-sectional, limiting the inferences that can be made about time- and practice-related effects. As such, future work should focus on conducting longitudinal studies that provide a more valid and rigorous test of the early choice hypothesis to explain the dyslexia and creativity relationship.

The current meta-analysis is limited by the small number of studies available for inclusion in this paper. The small number of available studies, along with the large heterogeneity of the available studies, can pose limits on estimating a reliable effect size. For example, it is possible that different creativity tasks tap into different processes and may involve several discrete abilities concurrently, such as verbal versus non-verbal skills in addition to fluency versus originality abilities. Due to the small number of available studies, more targeted subgroup analyses and complex moderation analyses, such as with multiple characteristics concurrently, were not possible. Additionally, the small number of available papers limits the number of different populations available for inclusion, resulting in potentially low generalizability of the results. For example, of the available studies, the majority of studies were conducted with European samples, with only three samples from Israel (Kasirer & Mashal, 2017) and one sample from Brazil (Alves &

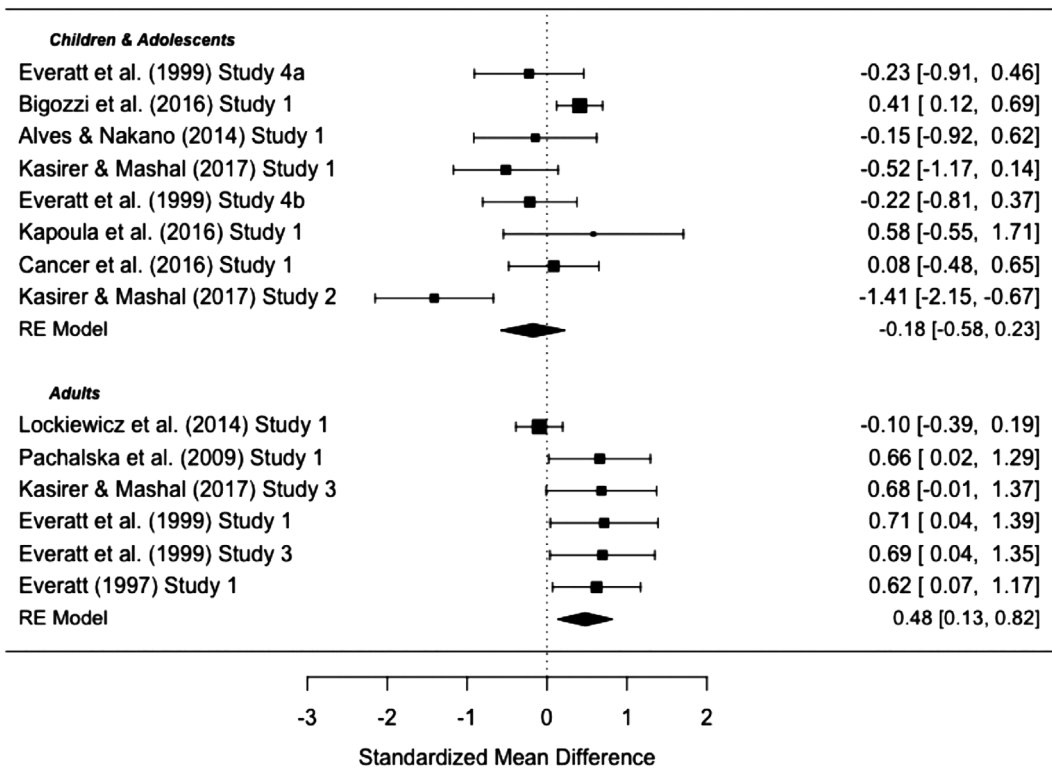


FIGURE 5 Effect sizes and confidence intervals for the two age groups

TABLE 3 Analysis of continuous moderators of the dyslexia–creativity relation

Moderator	N	k	b	SE _b	95% CI _{LB}	95% CI _{UB}
Intelligence	527	10	0.79	0.55	-0.28	1.87
Mean age	736	12	0.05	0.03	-0.0039	0.10
Proportion of females	758	12	1.85	1.74	-1.56	5.26

Abbreviations: *b*, Slope in the moderation model; *k*, Number of samples; *N*, Sample size; SE_{*b*}, Standard error of the slope; 95% CI_{LB} and CI_{UB} = 95% Confidence interval.

de Cássia Nakano, 2014). This suggests that more studies from other regions are needed to allow a more comprehensive and reliable synthesis of the literature.

Future studies should also look into the effect of different types of written scripts on the dyslexia–creativity relationship. There are two main types of scripts: alphabetic and logographic (Liu, 1995). Alphabetic scripts are scripts that rely on individual letters of an alphabet, such as those used in English and related languages, while logographic scripts are scripts that rely on whole-word or whole-syllable symbols, such as those used in Chinese languages (Sampson, 1985). In the current study, the dyslexia–creativity relationship was only studied in samples using alphabetic scripts, even though developmental dyslexia has been found to be equally prevalent in samples, which use logographic scripts and in samples, which use alphabetic scripts (Stevenson et al., 1982). In neuroimaging research, the brains of Chinese-speaking individuals with developmental dyslexia have been found to be structurally and functionally different from those of English-speaking individuals with developmental dyslexia (Siok, Niu, Jin, Perfetti, & Tan, 2008). This is important as the mechanism underlying dyslexia may differ in these populations, due to the

neurobiological nature of dyslexia (Yang, Yang, Chen, Zhang, & Bi, 2016). If the neurobiological *compensation hypothesis* is true, this difference in brain structure and function may potentially lead to different effects on creativity. Hence, the type of script used by the individual can be a promising moderator of the dyslexia-creativity relationship and should be looked into in future research.

Future studies should also look into the effect of socio-economic status (SES) on the dyslexia-creativity relationship. Due to the lack of available data, SES was not examined as a potential moderator in the current meta-analysis. In other research, SES and creativity have been shown to be positively correlated (Dai et al., 2012; Lichtenwalner & Maxwell, 1969). As such, between groups not matched on SES, results could be skewed in favour of the group with higher SES. Additionally, low SES individuals with learning disabilities, including dyslexia, are usually more impaired by their disability (Vernon, 1999) or are impaired in different ways compared to high SES individuals with the same learning disabilities (Oliver, Sapey, & Thomas, 2012). Therefore, even within groups with dyslexia, the effect of dyslexia on creativity may vary as a function of SES.

5 | CONCLUSION

Overall, the current meta-analysis found limited support for the idea that individuals with dyslexia are more creative than those without dyslexia. This was the case regardless of differences in critical method factors, such as the type of creativity task used, the type of creativity outcome assessed, and whether intelligence was matched, as well as gender. Nonetheless, the only significant effect size observed was in the adult group, where adults with dyslexia significantly outperformed controls, but not in the group with young children. This provides preliminary evidence for the early choice hypothesis, although more longitudinal examinations in future will be needed to ascertain this explanation.

ACKNOWLEDGMENTS

The authors received no financial support for the research, authorship and/or publication of this article. We are grateful to Kenneth Tan and Phyu Zin Htet for helpful comments on an earlier version of the manuscript, to Rebecca Maniates for her assistance with literature retrieval, and to Lucas Ethan Tiong, Manmeet Kaur and Meenakshi Kothari for their assistance with data coding.

ORCID

Nadyanna M. Majeed  <https://orcid.org/0000-0003-0148-0846>

Jacinth J. X. Tan  <https://orcid.org/0000-0001-6058-0076>

REFERENCES

References marked with an asterisk (*) indicate studies included in the review

- Abraham, A., Pieritz, K., Thybusch, K., Rutter, B., Kröger, S., Schweckendiek, J., ... Hermann, C. (2012). Creativity and the brain: Uncovering the neural signature of conceptual expansion. *Neuropsychologia*, 50(8), 1906–1917. <https://doi.org/10.1016/j.neuropsychologia.2012.04.015>
- Alexander-Passe, N. (2015). The dyslexia experience: Difference, disclosure, labelling, discrimination and stigma. *Asia Pacific Journal of Developmental Differences*, 2, 202–233. <https://doi.org/10.3850/S2345734115000290>
- *Alves, R. J. R., & de Cássia Nakano, T. (2014). Creativity and intelligence in children with and without developmental dyslexia. *Paidéia (Ribeirão Preto)*, 24(59), 361–369. <https://doi.org/10.1590/1982-43272459201410>
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Arlington, VA: American Psychiatric Association. Retrieved from <https://doi.org/10.1176/appi.books.9780890425596>
- Antonietti, A., Giorgetti, M., & Pizzingrilli, P. (2011). *Io penso creativo. Valutare e potenziare gli aspetti creativi del pensiero [I think creative. Assess and improve creative aspects of thinking]*. Firenze, Italy: Giunti.

- Benedek, M., Jauk, E., Sommer, M., Arendasy, M., & Neubauer, A. C. (2014). Intelligence, creativity, and cognitive control: The common and differential involvement of executive functions in intelligence and creativity. *Intelligence*, 46, 73–83. <https://doi.org/10.1016/j.intell.2014.05.007>
- *Bigozzi, L., Tarchi, C., Pinto, G., & Donfrancesco, R. (2016). Divergent thinking in Italian students with and without reading impairments. *International Journal of Disability, Development and Education*, 63(4), 450–466. <https://doi.org/10.1080/1034912X.2015.1127335>
- Büttner, G., & Hasselhorn, M. (2011). Learning disabilities: Debates on definitions, causes, subtypes, and responses. *International Journal of Disability, Development and Education*, 58(1), 75–87. <https://doi.org/10.1080/1034912X.2011.548476>
- *Cancer, A., Manzoli, S., & Antonietti, A. (2016). The alleged link between creativity and dyslexia: Identifying the specific process in which dyslexic students excel. *Cogent Psychology*, 3(1), 1190309. <https://doi.org/10.1080/23311908.2016.1190309>
- Chakravarty, A. (2009). Artistic talent in dyslexia—A hypothesis. *Medical Hypotheses*, 73(4), 569–571. <https://doi.org/10.1016/j.mehy.2009.05.034>
- Cockcroft, K., & Hartgill, M. (2004). Focusing on the abilities in learning disabilities: Dyslexia and creativity. *Education as Change*, 8(1), 61–79. <https://doi.org/10.1080/16823200409487081>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hillsdale, NJ: L. Erlbaum Associates.
- Craggs, J., Sanchez, J., Kibby, M., Gilger, J., & Hynd, G. (2006). Brain morphology and neuropsychological profiles in a family displaying dyslexia and superior nonverbal intelligence. *Cortex*, 42(8), 1107–1118. [https://doi.org/10.1016/S0010-9452\(08\)70222-3](https://doi.org/10.1016/S0010-9452(08)70222-3)
- Dai, D. Y., Tan, X., Marathe, D., Valtcheva, A., Pruzek, R. M., & Shen, J. (2012). Influences of social and educational environments on creativity during adolescence: Does SES matter? *Creativity Research Journal*, 24(2–3), 191–199. <https://doi.org/10.1080/10400419.2012.677338>
- Das, J. P., Mishra, R. K., & Kirby, J. R. (1994). Cognitive patterns of children with dyslexia: A comparison between groups with high and average nonverbal intelligence. *Journal of Learning Disabilities*, 27(4), 235–242. <https://doi.org/10.1177/002221949402700405>
- Duffy, F. H., Denckla, M. B., Bartels, P. H., & Sandini, G. (1980). Dyslexia: Regional differences in brain electrical activity by topographic mapping. *Annals of Neurology*, 7(5), 412–420. <https://doi.org/10.1002/ana.410070505>
- Eisen, M. L. (1989). Assessing differences in children with learning disabilities and normally achieving students with a new measure of creativity. *Journal of Learning Disabilities*, 22(7), 462–464. <https://doi.org/10.1177/002221948902200715>
- *Everatt, J. (1997). The abilities and disabilities associated with adult developmental dyslexia. *Journal of Research in Reading*, 20(1), 13–21. <https://doi.org/10.1111/1467-9817.00016>
- *Everatt, J., Steffert, B., & Smythe, I. (1999). An eye for the unusual: Creative thinking in dyslexics. *Dyslexia*, 5(1), 28–46. [https://doi.org/10.1002/\(SICI\)1099-0909\(199903\)5:1<28::AID-DYS126>3.0.CO;2-K](https://doi.org/10.1002/(SICI)1099-0909(199903)5:1<28::AID-DYS126>3.0.CO;2-K)
- Finke, R. A., Ward, T. B., & Smith, S. M. (1996). *Creative cognition: Theory, research, and applications*. Cambridge, MA: MIT Press.
- Furnham, A., Batey, M., Anand, K., & Manfield, J. (2008). Personality, hypomania, intelligence and creativity. *Personality and Individual Differences*, 44(5), 1060–1069. <https://doi.org/10.1016/j.paid.2007.10.035>
- Galaburda, A. M., Sherman, G. F., Rosen, G. D., Aboitiz, F., & Geschwind, N. (1985). Developmental dyslexia: Four consecutive patients with cortical anomalies. *Annals of Neurology*, 18(2), 222–233. <https://doi.org/10.1002/ana.410180210>
- Ganttt, L. M., & Anderson, F. (2009). The Formal Elements Art Therapy Scale: A measurement system for global variables in art. *Art Therapy*, 26(3), 124–129. <https://doi.org/10.1080/07421656.2009.10129372>
- Geschwind, N., & Galaburda, A. M. (1987). *Cerebral lateralization: Biological mechanisms, associations, and pathology*. Cambridge, MA: MIT Press.
- Guilford, J. P., & Hoepfner, R. (1971). *The analysis of intelligence*. New York, NY: McGraw-Hill.
- Gunnel Ingesson, S. (2006). Stability of IQ measures in teenagers and young adults with developmental dyslexia. *Dyslexia*, 12(2), 81–95. <https://doi.org/10.1002/dys.306>
- Gunnel Ingesson, S. (2007). Growing up with dyslexia: Interviews with teenagers and young adults. *School Psychology International*, 28(5), 574–591. <https://doi.org/10.1177/0143034307085659>
- Hedges, L. V. (1981). Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, 6(2), 107–128. <https://doi.org/10.3102/10769986006002107>
- Jeffries, S., & Everatt, J. (2004). Working memory: Its role in dyslexia and other specific learning difficulties. *Dyslexia*, 10(3), 196–214. <https://doi.org/10.1002/dys.278>
- *Kapoula, Z., Ruiz, S., Spector, L., Mocerovi, M., Gaertner, C., Quilici, C., & Vernet, M. (2016). Education influences creativity in dyslexic and non-dyslexic children and teenagers. *PLoS ONE*, 11(3), e0150421. <https://doi.org/10.1371/journal.pone.0150421>
- *Kasirer, A., & Mashal, N. (2017). Comprehension and generation of metaphoric language in children, adolescents, and adults with dyslexia: Metaphor comprehension and generation in dyslexia. *Dyslexia*, 23(2), 99–118. <https://doi.org/10.1002/dys.1550>

- Kendall, M. G. (1938). A new measure of rank correlation. *Biometrika*, 30(1–2), 81–93. <https://doi.org/10.2307/2332226>
- Khalil, R., Godde, B., & Karim, A. A. (2019). The link between creativity, cognition, and creative drives and underlying neural mechanisms. *Frontiers in Neural Circuits*, 13, 18. <https://doi.org/10.3389/fncir.2019.00018>
- Lee, H., & Kim, K. H. (2011). Can speaking more languages enhance your creativity? Relationship between bilingualism and creative potential among Korean American students with multicultural link. *Personality and Individual Differences*, 50(8), 1186–1190. <https://doi.org/10.1016/j.paid.2011.01.039>
- Lerner, J. W. (1989). Educational interventions in learning disabilities. *Journal of the American Academy of Child & Adolescent Psychiatry*, 28(3), 326–331. <https://doi.org/10.1097/00004583-198905000-00004>
- Lichtenwalner, J. S., & Maxwell, J. W. (1969). The relationship of birth order and socio-economic status to the creativity of preschool children. *Child Development*, 40(4), 1241–1247. <https://doi.org/10.2307/1127028>
- Liu, I.-M. (1995). Script factors that affect literacy: Alphabetic vs. logographic languages. In I. Taylor & D. R. Olson (Eds.), *Scripts and literacy* (pp. 145–162). Amsterdam, the Amsterdam: Kluwer Academic.
- *Łockiewicz, M., Bogdanowicz, K. M., & Bogdanowicz, M. (2014). Psychological resources of adults with developmental dyslexia. *Journal of Learning Disabilities*, 47(6), 543–555. <https://doi.org/10.1177/0022219413478663>
- Lum, J. A. G., Ullman, M. T., & Conti-Ramsden, G. (2013). Procedural learning is impaired in dyslexia: Evidence from a meta-analysis of serial reaction time studies. *Research in Developmental Disabilities*, 34(10), 3460–3476. <https://doi.org/10.1016/j.ridd.2013.07.017>
- Lyon, G. R., Shaywitz, S. E., & Shaywitz, B. A. (2003). A definition of dyslexia. *Annals of Dyslexia*, 53(1), 1–14. <https://doi.org/10.1007/s11881-003-0001-9>
- Matczak, A., Jaworowska, A., & Stańczak, J. (2000). TCT-DP. Rysunkowy Test Twórczego Myślenia K. K. Urbana i H. G. Jellena. Podręcznik [TCT-DP. K. K. Urban & H. G. Jellen's Test for Creative Thinking Drawing Production: Manual]. Warsaw, Poland: Pracownia Testów Psychologicznych Polskiego Towarzystwa Psychologicznego.
- Metcalfe, J. (1986). Feeling of knowing in memory and problem solving. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 12(2), 288–294. <https://doi.org/10.1037/0278-7393.12.2.288>
- Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G., & PRISMA Group. (2009). Reprint—Preferred Reporting Items for systematic reviews and meta-analyses: The PRISMA statement. *Physical Therapy*, 89(9), 873–880. <https://doi.org/10.1093/ptj/89.9.873>
- Morris, D. K., & Turnbull, P. A. (2007). The disclosure of dyslexia in clinical practice: Experiences of student nurses in the United Kingdom. *Nurse Education Today*, 27(1), 35–42. <https://doi.org/10.1016/j.nedt.2006.01.017>
- Nakano, T. d. C., Wechsler, S. M., & Primi, R. (2011). *Teste de criatividade figural infantil*. São Paulo, Brazil: Vetor.
- O'Brien, B. A., Wolf, M., & Lovett, M. W. (2012). A taxometric investigation of developmental dyslexia subtypes: Taxometric investigation of dyslexia subtypes. *Dyslexia*, 18(1), 16–39. <https://doi.org/10.1002/dys.1431>
- O'Hare, A. (2010). Dyslexia: What do paediatricians need to know? *Paediatrics and Child Health*, 20(7), 338–343. <https://doi.org/10.1016/j.paed.2010.04.004>
- Oliver, M., Sapey, B., & Thomas, P. (2012). *Social work with disabled people* (4th ed.). New York, NY: Palgrave Macmillan.
- *Pachalska, M., Bogdanowicz, K., Tomaszewska, K., Łockiewicz, M., & Bogdanowicz, M. (2009). The stimulation of creative activity in dyslexic adults. *Acta Neuropsychologica*, 7(2), 113–130.
- Paulesu, E. (2001). Dyslexia: Cultural diversity and biological unity. *Science*, 291(5511), 2165–2167. <https://doi.org/10.1126/science.1057179>
- Peterson, R. L., & Pennington, B. F. (2012). Developmental dyslexia. *The Lancet*, 379(9830), 1997–2007. [https://doi.org/10.1016/S0140-6736\(12\)60198-6](https://doi.org/10.1016/S0140-6736(12)60198-6)
- Plucker, J. A., & Makel, M. C. (2010). Assessment of creativity. In J. C. Kaufman & R. J. Sternberg (Eds.), *The Cambridge handbook of creativity* (pp. 48–73). Cambridge, England: Cambridge University Press. Retrieved from <https://doi.org/10.1017/CBO9780511763205.005>
- Preckel, F., Holling, H., & Wiese, M. (2006). Relationship of intelligence and creativity in gifted and non-gifted students: An investigation of threshold theory. *Personality and Individual Differences*, 40(1), 159–170. <https://doi.org/10.1016/j.paid.2005.06.022>
- Ramus, F. (2004). Neurobiology of dyslexia: A reinterpretation of the data. *Trends in Neurosciences*, 27(12), 720–726. <https://doi.org/10.1016/j.tins.2004.10.004>
- R Core Team. (2020). *R: A language and environment for statistical computing*, Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>
- Ridley, D. R., & Birney, R. C. (1967). Effects of training procedures on creativity test scores. *Journal of Educational Psychology*, 58(3), 158–164. <https://doi.org/10.1037/h0024655>
- Ritchie, S. J., Luciano, M., Hansell, N. K., Wright, M. J., & Bates, T. C. (2013). The relationship of reading ability to creativity: Positive, not negative associations. *Learning and Individual Differences*, 26, 171–176. <https://doi.org/10.1016/j.lindif.2013.02.009>
- Sampson, G. (1985). *Writing systems: A linguistic introduction (Original printing)*. Stanford, CA: Stanford Univ. Press.

- Schubert, D. S. P. (1973). Intelligence as necessary but not sufficient for creativity. *The Journal of Genetic Psychology*, 122(1), 45–47. <https://doi.org/10.1080/00221325.1973.10533169>
- Shaywitz, B. A., Shaywitz, S. E., Pugh, K. R., Mencl, W. E., Fulbright, R. K., Skudlarski, P., ... Gore, J. C. (2002). Disruption of posterior brain systems for reading in children with developmental dyslexia. *Biological Psychiatry*, 52(2), 101–110. [https://doi.org/10.1016/S0006-3223\(02\)01365-3](https://doi.org/10.1016/S0006-3223(02)01365-3)
- Shaywitz, S. E., & Shaywitz, B. A. (2005). Dyslexia (specific reading disability). *Biological Psychiatry*, 57(11), 1301–1309. <https://doi.org/10.1016/j.biopsych.2005.01.043>
- Silvia, P. J. (2015). Intelligence and creativity are pretty similar after all. *Educational Psychology Review*, 27(4), 599–606. <https://doi.org/10.1007/s10648-015-9299-1>
- Siok, W. T., Niu, Z., Jin, Z., Perfetti, C. A., & Tan, L. H. (2008). A structural-functional basis for dyslexia in the cortex of Chinese readers. *Proceedings of the National Academy of Sciences of the United States of America*, 105(14), 5561–5566. <https://doi.org/10.1073/pnas.0801750105>
- Sterne, J. A. C., & Egger, M. (2001). Funnel plots for detecting bias in meta-analysis. *Journal of Clinical Epidemiology*, 54(10), 1046–1055. [https://doi.org/10.1016/S0895-4356\(01\)00377-8](https://doi.org/10.1016/S0895-4356(01)00377-8)
- Stevenson, H. W., Stigler, J. W., Lucker, G. W., Lee, S., Hsu, C., & Kitamura, S. (1982). Reading disabilities: The case of Chinese, Japanese, and English. *Child Development*, 53(5), 1164–1181.
- Sturgell, I. (2008). Touchstone texts: Fertile ground for creativity. *The Reading Teacher*, 61(5), 411–414. <https://doi.org/10.1598/RT.61.5.5>
- Tafti, M. A., Hameedy, M. A., & Baghal, N. M. (2009). Dyslexia, a deficit or a difference: Comparing the creativity and memory skills of dyslexic and nondyslexic students in Iran. *Social Behavior and Personality: An International Journal*, 37(8), 1009–1016. <https://doi.org/10.2224/sbp.2009.37.8.1009>
- Tan, C.-S., & Qu, L. (2015). Stability of the positive mood effect on creativity when task switching, practice effect, and test item differences are taken into consideration. *The Journal of Creative Behavior*, 49(2), 94–110. <https://doi.org/10.1002/jocb.56>
- Tarver, S. G., Ellsworth, P. S., & Rounds, D. J. (1980). Figural and verbal creativity in learning disabled and nondisabled children. *Learning Disability Quarterly*, 3(3), 11–18. <https://doi.org/10.2307/1510626>
- Thomson, M. (2003). Monitoring dyslexics' intelligence and attainments: A follow-up study. *Dyslexia*, 9(1), 3–17. <https://doi.org/10.1002/dys.232>
- Torrance, E. P. (1966). *Torrance tests of creative thinking. Norms-technical manual. Research edition. Verbal tests, forms A and B. Figural tests, forms A and B*. Princeton, NJ: Personnel Press.
- van Bergen, E., de Jong, P. F., Maassen, B., Krikhaar, E., Plakas, A., & van der Leij, A. (2014). IQ of four-year-olds who go on to develop dyslexia. *Journal of Learning Disabilities*, 47(5), 475–484. <https://doi.org/10.1177/0022219413479673>
- Vernon, A. (1999). The dialectics of multiple identities and the disabled people's movement. *Disability & Society*, 14(3), 385–398. <https://doi.org/10.1080/09687599926217>
- Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*, 36(3), 1–48. <https://doi.org/10.18637/jss.v036.i03>
- von Károlyi, C., Winner, E., Gray, W., & Sherman, G. F. (2003). Dyslexia linked to talent: Global visual-spatial ability. *Brain and Language*, 85(3), 427–431. [https://doi.org/10.1016/S0093-934X\(03\)00052-X](https://doi.org/10.1016/S0093-934X(03)00052-X)
- Wagner, R. K., Zirps, F. A., Edwards, A. A., Wood, S. G., Joyner, R. E., Becker, B. J., ... Beal, B. (2020). The prevalence of dyslexia: A new approach to its estimation. *Journal of Learning Disabilities*, 53(5), 354–365. <https://doi.org/10.1177/0022219420920377>
- Wallach, M. A. (1970). Creativity. In P. H. Mussen (Ed.), *Carmichael's manual of child psychology* (3rd ed., pp. 1211–1272). New York, NY: Wiley.
- Wang, A. Y. (2012). Exploring the relationship of creative thinking to reading and writing. *Thinking Skills and Creativity*, 7(1), 38–47. <https://doi.org/10.1016/j.tsc.2011.09.001>
- Weisberg, R. W., & Alba, J. W. (1981). An examination of the alleged role of “fixation” in the solution of several “insight” problems. *Journal of Experimental Psychology: General*, 110(2), 169–192. <https://doi.org/10.1037/0096-3445.110.2.169>
- Whitehouse, C. C. (1983). Analysis of WISC-R coding performance of normal and dyslexic readers. *Perceptual and Motor Skills*, 57(3), 951–960. <https://doi.org/10.2466/pms.1983.57.3.951>
- Williams, F. (1993). *CAP: Creativity assessment packet*. Austin, TX: Pro-Ed.
- Wimmer, H. (1993). Characteristics of developmental dyslexia in a regular writing system. *Applied Psycholinguistics*, 14(1), 1–33. <https://doi.org/10.1017/S0142716400010122>
- Winner, E., von Károlyi, C., Malinsky, D., French, L., Seliger, C., Ross, E., & Weber, C. (2001). Dyslexia and visual-spatial talents: Compensation vs deficit model. *Brain and Language*, 76(2), 81–110. <https://doi.org/10.1006/brln.2000.2392>
- Yang, Y.-H., Yang, Y., Chen, B.-G., Zhang, Y.-W., & Bi, H.-Y. (2016). Anomalous cerebellar anatomy in Chinese children with dyslexia. *Frontiers in Psychology*, 7(324), 1–9. <https://doi.org/10.3389/fpsyg.2016.00324>

- Yap, R. L., & van der Leij, A. (1994). Testing the automatization deficit hypothesis of dyslexia via a dual-task paradigm. *Journal of Learning Disabilities*, 27(10), 660–665. <https://doi.org/10.1177/002221949402701006>
- Ziegler, J. C., Perry, C., Ma-Wyatt, A., Ladner, D., & Schulte-Körne, G. (2003). Developmental dyslexia in different languages: Language-specific or universal? *Journal of Experimental Child Psychology*, 86(3), 169–193. [https://doi.org/10.1016/S0022-0965\(03\)00139-5](https://doi.org/10.1016/S0022-0965(03)00139-5)

How to cite this article: Majeed NM, Hartanto A, Tan JJX. Developmental dyslexia and creativity: A meta-analysis. *Dyslexia*. 2021;27:187–203. <https://doi.org/10.1002/dys.1677>